

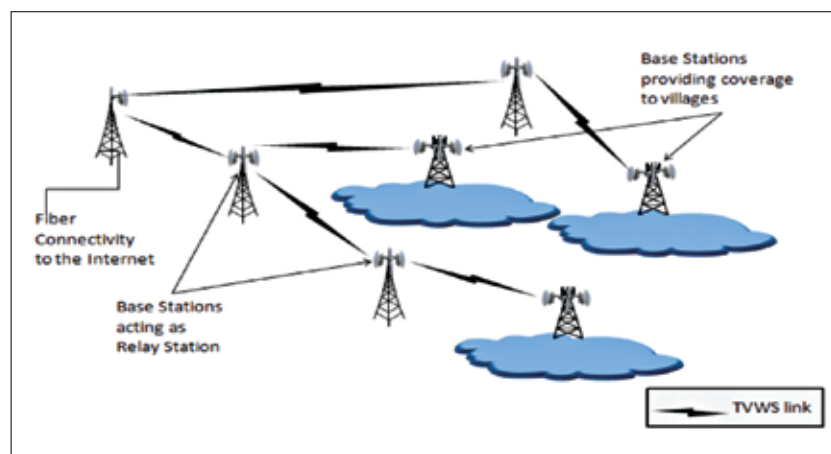
Additionally, there were also white spaces, unused spectrum to differentiate between two TV channels. Making these available for mobile connectivity has proved a valuable source of additional capacity worldwide. Additionally, the spectrum that has been acquired by operators has been in small chunks at different points in time. There is a need to start with a clean slate, and allocate contiguous spectrum. There can also be mechanisms implemented that ensure that the spectrum is used most effectively. These include making the spectrum available for defence purposes only when needed, and having frequent, periodic and transparent auctions. Another important thing that the government needs to do is publish a roadmap for how and when spectrum will be made available in the future, which would allow the service providers to build up their infrastructure appropriately. The rollout is slowed down because of a lack of clarity on the future availability of spectrum.

Now, an interesting solution to many of these problems can come from an unexpected direction – that of small cable TV operators across the country. From the very beginning, these operators have been using all kinds of *jugaad* to ensure that people get access to television. They have also been using indigenously manufactured technology for realising the same. They have the best on-ground knowledge, and can easily adapt their existing systems to provide last mile broadband connectivity to the underserved. They can circumvent problems that ISPs have in laying optical fibres, and TSPs have in setting up telecom infrastructure, which is perceived to be dangerous, because of irrational fears of radiation. Many cable operators around the country are, in fact, already delivering broadband connectivity, with reasonably-priced plans. Here, the small sizes of the operations is an advantage. In fact, there have been plans by the government to use cable operators for delivering broadband since 1999, but no concrete plans have been implemented yet.

Irrespective of whether the last mile connectivity is wired or wireless, the connectivity in both forms can be bottlenecked if there is a lack of capacity in backhaul networks. WiFi hotspots in

large metropolitan cities is a proposal for providing wireless connectivity without the need for telecom infrastructure. However, such hotspots would need additional capacity in the backhaul networks. Conversely, in the villages, especially the small ones, all the broadband services would by necessity have to be provided through telecom infrastructure, and not optic fibre. This is because there are over seven lakh small settlements, at a distance of at least two kilometers from each other, with a population of

of the population having access to LTE services, only 8 percent of the population are actually using it. According to the latest State of the Information Society by the International Telecommunications Union (ITU), 63 percent of mobile owners who choose not to subscribe to the internet say that they do not know what the internet is for. Another 23 percent know what the internet is for, but do not intend to ever access it on their mobile devices. The process to fix this starts at the school



Middle-mile wireless network

Credit: IITB

less than five hundred. Providing access to these locations would also require a robust, flexible backhaul network. Researchers at IITB have proposed a single, elegant solution that can improve broadband experience across India, for both rural and urban population, without really disturbing any existing deployments. In urban areas, the available spectrum is heavily used because of the dense population. However, in rural areas, the spectrum is not heavily used, which is about 56.27 percent of the area of the country. The idea is to setup a multi-hop wireless mesh network that acts as a backhaul network in the rural areas of the population. This provides a backhaul for the urban centres, and at the same time extends connectivity to rural areas. Through license sharing agreements with TV broadcasters, such a system could be setup without a necessity of the broadcasters to vacate their allocated spectrum.

Even where broadband services are easily available, adoption is low. Digital literacy is perhaps the biggest roadblock here. This is because despite 70 percent

level itself, and the National Computer Literacy Drive is a step in this direction. However, it is unfair to expect everyone to just learn English or Hindi as their first language. Merely translating content into local language is not enough, there needs to be local content that is culturally relevant to the consumers, for an increase in adoption of digital technologies. This is one of the reasons that TikTok has proved to be so successful in India. There is little pressure for rollout of connectivity infrastructure to the rural areas, because India has just about the lowest ARPUs (average revenue per user) in the world, at less than \$2 per customer. The fact that people have tried to pay the developers of Ludo King with demand drafts, is proof that this is not an unwillingness of people from paying for content, rather it is the lack of availability of such content. Without relevant content, consumers simply do not know what to do with connectivity, and without connectivity, content creators do not have any incentive to create such content. The end user is suffering because of this unholy stalemate. ■